FINAL PROJECT PROPOSAL REPORT

ON

Global Sports Footwear - Retailer Analytics

ILS-Z 511 Database Design

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01 Community / Population

This project focuses on sports footwear retailers operating in the global marketplace. These are regional, multi-brand retail chains and independent sports goods stores - that must make continuous, data-driven decisions about pricing, product assortment, and inventory allocation across diverse markets worldwide.

1.1. Defining the Community

We will be focusing on Retailers as our primary community :

- Retailers - Retail chains, independent sports stores, and online footwear platforms that determine in-store and online promotional pricing on a market-by-market basis, manage inventory assortment across categories (Running, Training, Basketball, Lifestyle, Gym), and identify which customer income segments drive the most revenue in each sales channel.

Retailers are treated as a community in this project because the decisions made by brands - setting base price and authorised discount ceilings directly constrain and inform the decisions made by retailers - promotional depth, channel-specific pricing. The dataset captures outcomes from both perspectives in a single transactional record, making it possible to analyse the full pricing-to-purchase chain in one place.

1.2. Why We Chose This Community

The choice of retailers as the focal community was driven by three key considerations:

Data Alignment - The dataset is structured around commercial transactions, not individual consumer journeys. Every row records a sale from the seller's perspective - capturing price set, discount applied, channel used, revenue generated, and units moved. This is the language of retail and brand operations.

Decision-Making Authority - Both problem statements (pricing strategy and customer behaviour) are directly actionable only by organisations with the power to change prices and stock decisions. Brands and retailers hold that authority; individual consumers do not.

Commercial Scale - The dataset spans six major global brands, multiple countries, and eight years (2018-2026). This scale mirrors the multi-market reporting environment that brand managers and retail buyers actually operate in, making the analysis realistic and applicable.

02 Dataset

2.1 Dataset Overview

The dataset used in this project is the **Global Sports Footwear Sales & Consumer Behavior Dataset**, publicly available on Kaggle at the following URL:

Source: https://www.kaggle.com/datasets/aliiihussain/sports-footwear-sales-and-consumer-behavior/data **Published by:** Alii Hussain (Kaggle contributor) **License:** Open Data - publicly accessible for educational and research use

This dataset was selected because it directly aligns with the two problem statements defined for this project - pricing strategy analysis and customer behaviour profiling that provides all the attributes necessary to populate the normalised relational database schema we have designed.

2.2 Dataset Characteristics

The dataset is a **realistic simulation** of global sports footwear sales transactions spanning an eight-year period. It is designed to closely model real-world commercial sales patterns without containing any personally identifiable consumer data.

Property	Detail
Total Records	30,000 rows
Total Attributes	18 columns
Time Period Covered	December 31, 2017 - December 30, 2026
Mean Transaction Date	June 28, 2022
Brands Covered	6 (Nike, Adidas, Puma, Reebok, ASICS, New Balance)
Countries Covered	6 (UAE, UK, USA, India, Germany, Pakistan)
Data Quality	100% valid across all 18 columns - 0 missing, 0 mismatched values

Table 1. Dataset Characteristics

The dataset is entirely clean, with no preprocessing required before loading into a relational database. Every column across all 30,000 rows returned 100% valid values with zero missing entries and zero type mismatches, as confirmed by the data profiling reports.

2.3 Description

Attribute	Description
order_id	Unique ID for each sale ID
order_date	Sale date
brand	Brand of the shoe
model_name	Specific model of the shoe
category	Running, Training, Basketball, Lifestyle, Gym
gender	Men, Women, Unisex
size	Shoe size
color	Color of the shoe
base_price_usd	Original price
discount_percent	Discount applied (%)
final_price_usd	Price after discount
units_sold	Quantity sold
revenue_usd	Total revenue for the order
payment_method	Card, Cash, Wallet, Bank Transfer
sales_channel	Online / Retail Store
country	Country of sale
customer_income_level	Low / Medium / High
customer_rating	Average rating from 3.0 to 5.0

Table 2: Dataset Description

2.4 Proposed ER Diagram

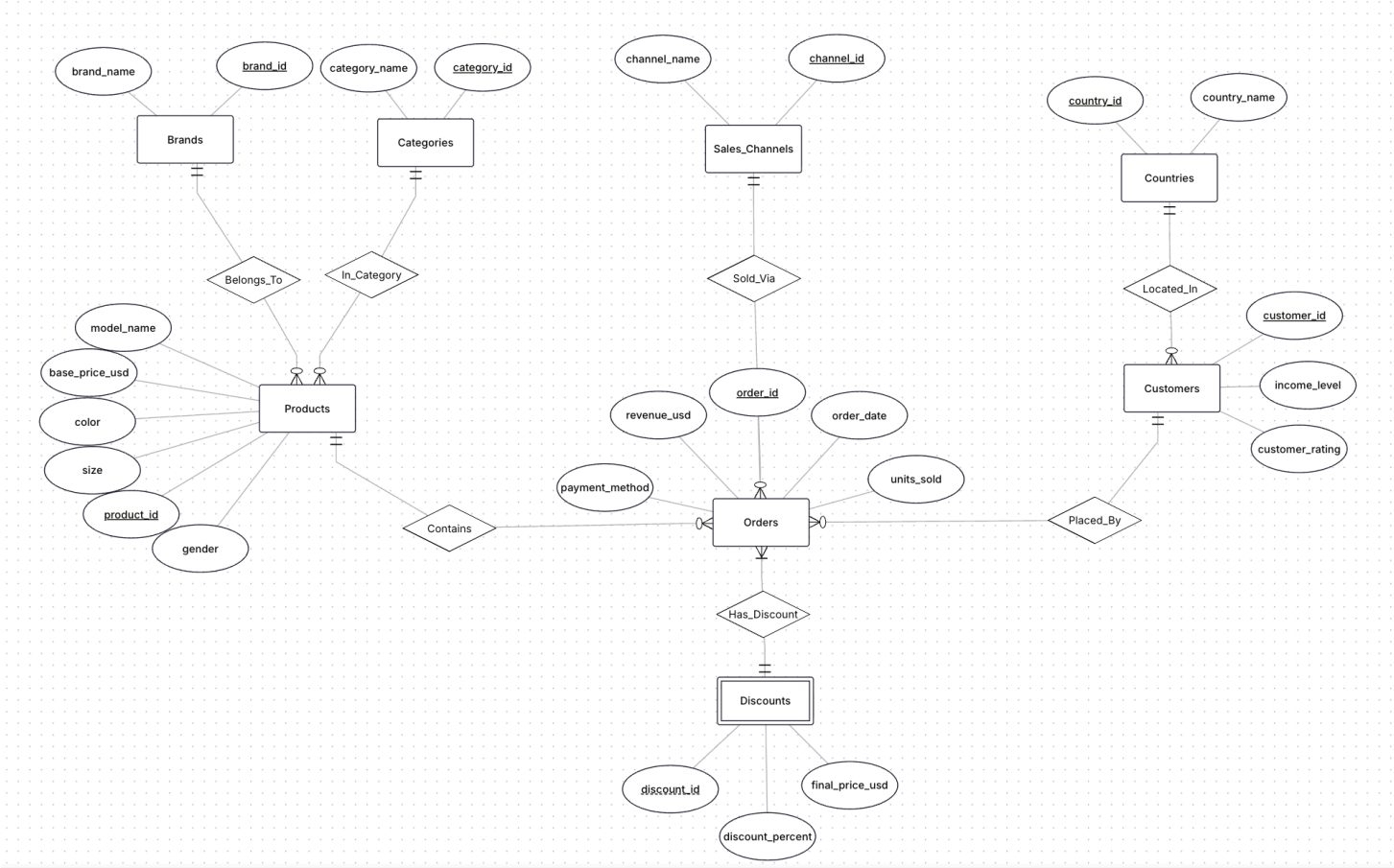


Figure 1: Proposed ER Diagram

03 Problems the Dataset Allows the Community to Solve

The Global Sports Footwear Sales dataset provides us with a rich, multi-dimensional view of commercial performance. It uniquely combines pricing variables, sales outcomes, customer demographics, and satisfaction ratings in a single flat file - a combination that, once migrated into a normalised relational database, enables structured answers to two core operational problems facing this community.

3.1. How the Dataset Relates to the Community

The eighteen attributes in the raw dataset map precisely onto the decision variables and performance metrics that brands and retailers use day-to-day:

Dataset Attribute	What It Represents	How Retailers Use It
base_price_usd / final_price_usd	Price set by brand and price paid after discount	Evaluate pricing power and discount margin erosion per brand
discount_percent	Promotional depth applied at point of sale	Determine effective discount thresholds by category
units_sold / revenue_usd	Volume and value outcome of each transaction	Core KPIs for sales performance measurement
brand / category	Product classification dimensions	Assortment planning and brand-level investment decisions
sales_channel	Online vs. physical retail	Channel-specific pricing and promotional strategy
customer_income_level	Low / Medium / High consumer income segment	Market segmentation and product-to-demographic matching
customer_rating	Satisfaction score (3.0–5.0) per purchase	Identify high-performing SKUs for inventory prioritisation
country / order_date	Geographic market and time dimension	Regional demand planning and seasonal trend analysis

Table 3: Dataset Community Relation Mappings

Crucially, no single attribute answers a retailer's question in isolation. It is the relational combination - pairing discount_percent with revenue_usd and brand, or joining customer_income_level with category and customer_rating - that produces actionable intelligence. This is precisely the motivation for migrating the flat dataset into a normalized relational database, so that multi-table joins can be executed cleanly and consistently.

3.2. How the Questions Were Chosen and How They Relate to the Community

The two problem statements were selected by matching the dataset's attributes to the two most pressing operational challenges faced by retail brand managers:

1. Pricing Strategy was chosen because pricing decisions - specifically how much to discount - represent the single most frequent and highest-stakes decision in retail operations. The dataset contains both the input variable (discount_percent) and the output metrics (units_sold, revenue_usd), making it possible to empirically test discount effectiveness rather than relying on intuition.

2. Customer Behaviour was chosen because retailers invest heavily in stocking decisions without sufficient visibility into which product combinations actually produce satisfied customers. The customer_rating column, when analysed alongside product attributes and demographic variables, provides exactly this visibility.

Both questions were also chosen because they are answerable at the aggregate level using transactional data, without requiring personally identifiable information - making them ethically appropriate for analysis on a dataset that models real consumer behavior patterns.

Problem 1: Pricing Strategy & Revenue Performance:

Sports footwear retailers operate on tight margins, yet many apply blanket discount policies without evidence of whether discounts drive incremental revenue or simply erode margin on sales that would have happened anyway. The dataset provides the complete pricing chain - base price, discount percentage, final price, revenue, and units - across 30,000+ transactions spanning six brands, five categories, and eight years, enabling a shift from assumption-based to data-supported pricing strategy.

1. **Discount Depth vs. Revenue Yield:** Group orders into discount tiers (0%, 1-10%, 11-20%, 21-30%, 30%+) and compare average and total revenue per tier to identify the inflection point where deeper discounts stop generating incremental revenue.

2. **Brand-Level Price Sensitivity:** Measure each brand's volume response across discount levels to determine which brands are discount-dependent and which hold volume at full price - informing promotional selection and negotiation with brand partners.

3. **Category-Level Discount Effectiveness:** Build a 2×2 matrix of discount sensitivity vs. full-price revenue performance by category to create a pricing playbook: which categories to discount aggressively, hold at price, or promote to clear inventory.

4. **Channel-Specific Pricing Behaviour:** Compare discount distributions and revenue outcomes between Online and Retail Store channels to determine whether promotional pricing should be concentrated digitally or spread across both.

Problem 2: Customer Behaviour & Product Preferences

Poor inventory planning - overstock or stockout is one of the most costly retailer failures. The challenge is predicting not just what sells, but what sells well in terms of both volume and customer satisfaction, a leading indicator of repeat purchase. The dataset connects product attributes with customer demographics and satisfaction ratings across a geographically diverse transaction set.

1. **Customer Loyalty:** Identify customer loyalty towards particular brands based on their previous purchases and their purchasing patterns regardless of the brand's pricing for newer models.

2. **Satisfaction Drivers by Product Attribute:** Compare the product profile (brand, category, gender, price, discount) of high-rated orders (≥4.5) versus low-rated ones to identify which attribute combinations consistently produce satisfied customers and should be prioritized in assortment planning.

3. **Geography & Demand Concentration:** Rank countries by revenue, order value, volume, and satisfaction, overlaid with income distribution, to surface commercially valuable markets and regions with unmet demand or satisfaction gaps.

4. **Payment Method as a Behavioural Signal:** Analyze order value, income level, channel preference, and satisfaction by payment method to build lightweight customer personas that inform promotional targeting - without requiring loyalty or identity data.

04 Ethics & Privacy Concerns

This project uses a synthetic dataset with no real personal identifiers. However, the analytical methodology - discount segmentation, income-level profiling, geographic demand ranking - is designed to be transferable to real transactional data. The ethical framework below therefore addresses both the current dataset and responsible real-world application.

4.1. Nature of the Dataset

The dataset (sourced from Kaggle) is synthetically generated, significantly reducing direct privacy risk. However, insights derived may approximate real-world distributions, the methodology is directly transferable to real data (requiring additional ethical oversight), and demographic proxies like income level and country represent sensitive characteristics when used in aggregate profiling.

4.2. Specific Privacy & Ethical Mitigations For in This Project

1. **Masking Sensitive Attributes:** In any real-world deployment of this schema, the customer_income_level and customer_rating fields should be stored in a restricted-access table with role-based access controls. Analysts working on pricing questions (Problem 1) do not require access to customer demographic data and should be provisioned accordingly. Column-level encryption or masking should be applied to these fields in production environments.

2. **No Personal Recommendations:** The analytical outputs of this project are intentionally designed to inform pricing policy and inventory strategy at the product and market level - not to generate personalised recommendations for individual consumers. The relational schema does not include a customer name, email, or unique identity field, and none will be added. This ensures the database cannot be repurposed for individual targeting without a deliberate and reviewable schema change.

3. **Aggregate-Only Reporting:** All sub-problem analyses (1a through 2d) produce group-level summaries - average revenue by discount tier, satisfaction scores by brand-category combination, revenue by country. No output presents data at the individual transaction level. This prevents re-identification and ensures compliance with data minimisation principles consistent with frameworks such as GDPR and CCPA.

4. **Restricted Use Scope:** The findings of this project are explicitly scoped to two legitimate retail use cases: (1) pricing strategy and revenue analysis, and (2) product assortment and inventory planning. Any extension of the methodology to dynamic personalised pricing, individual consumer credit scoring, or location-level surveillance would require separate ethical review and is outside the intended scope of this project.